Steve Herrin

Contact Information

jobs@steveherrin.com
650-814-8865
San Jose, CA

www.github.com/steveherrin
www.linkedin.com/in/herrinsteve

Summary

Engineering leader with over a decade of experience building and leading teams that use software, data, and machine learning to solve novel problems. Scientific (physics PhD) background with demonstrated adaptability to other fields like biotech.

Experience

Chan Zuckerberg Initiative, Redwood City, CA — August 2024 – present

Staff Software Engineer

- Developed a biological AI model benchmarking suite (`cz-benchmarks`), collaborating with NVIDIA, and integrated it into training pipelines
- Collaborated with AI researchers to productionize multiple embedding and generative models per quarter (using Run:ai and MLFlow), including transformers and diffusion models
- Built the identity and permissions (AuthZ/AuthN) system for the Virtual Cell Models platform using AWS Cognito, OpenFGA, Kubernetes, and Python (FastAPI) to manage fine-grained access to models, datasets, and admin functionality
- Augmented CI/CD with additional testing, type checking, linting, and other quality rules to accelerate automated development with coding agents

Pathos AI, Chicago, IL — June 2022 – July 2024

Vice President of Engineering

- Grew agile data-focused engineering team from zero to four engineers
- Developed a federated data catalog and management system using GCP, PostgreSQL, and Python to search for, manage, understand, and audit access to thousands of datasets and analysis results
- Prototyped a containerized large language model (LLM) retrieval-augmented generation (RAG) pipeline for scientific literature using pgvector and Google Cloud Platform
- Built analysis and processing environment from scratch using GCP and Terraform, and automated distributed scientific computing jobs using using Nextflow
- Developed Slack bots in Typescript/Deno to automate engineering and administrative tasks

D2G Oncology, Mountain View, CA — April 2021 – May 2022

Staff Software Engineer

- Created *in silico* simulations of PCR and DNA sequencing, used for oligo design, QC, and automated processing of multiplexed sequencing runs
- Built a pipeline using pydantic to automate ETL and validation of Benchling LIMS data from an HTTP API to a PostgreSQL warehouse
- Developed a Python library exposing GraphQL and REST APIs for accessing and traversing a large knowledge graph of lab, sequencing, public, and analysis data

23andMe, Sunnyvale, CA — January 2014 – March 2021

Engineering Individual Contributor (∼5 years; final title: sr. tech lead engineer)

- Architected and led building of a machine learning platform using Docker and AWS ECS to ensure quality, reproducibility, and rapid deployment of machine learning models, used for over 90% of production models
- Built a service to allow internal and external researchers to dynamically query k-anonymized data via Python and HBase for >5 million customers using a SQL interface, an HTTP API, or a web front end
- Migrated a 20 kLOC Django web application to the AWS cloud, upgrading the back-end from Python 2 to 3 and standardizing the front-end using React and Typescript
- Designed and implemented a Python library providing a unified API for accessing customer data across MySQL, HBase, and other data stores, eventually used for 100% of customer content
- Created Genotyping Services, a Django webapp on AWS allowing external researchers to easily run genomic studies, increasing sales by over 2% and leading to strategic data-sharing agreements

- Built 3 generations of distributed data pipelines with Celery, Luigi, and AWS Simple Workflow to run Python, C++, and R algorithms that impute, transform, and analyze petabytes of genetic data
- Developed and automated a maximum likelihood analysis combining private and public datasets that flagged $\sim 0.5\%$ of genotyping probes as bad for replacement in future platforms
- Automated genotype calling for SNPs and genes using a combination of unsupervised and supervised ML techniques in SKLearn

Engineering Management (∼2 years; title: engineering manager)

- Created, recruited for, and led 3 machine learning and data -focused engineering teams totaling 16 engineers, including a mix of leads, senior, and junior level individual contributors
- Formed and led team of engineering leads to standardize interviewing guidelines and open source release processes, subsequently adopted by all engineering teams

SLAC National Accelerator Lab, Menlo Park, CA — May 2008 – August 2013

Research Associate

- Applied machine learning algorithms & statistical analysis to improve detector energy resolution by 25%
- Repurposed the detector for 3D cosmic ray muon reconstruction using computer vision algorithms, yielding a 10x reduction in cosmogenic background uncertainty
- Created a PHP logbook webapp with a MySQL backend for tracking work on the EXO-200 experiment
- Built, networked, and programmed PLC control systems using over 600 channels of heterogeneous sensor data at a site with unreliable internet connectivity, successfully protecting $10M of cryogenic xenon
- Developed batch data pipelines using Python, C++, and shell scripts to routinely measure detector characteristics by processing TB of calibration data.
- Coordinated hardware and analysis software development with remote teams distributed around the world
- Mentored 1–2 junior graduate students (at any given time) on lab, coding, and statistical technique

Skills

Languages: Python, Rust, C, C++, SQL, Shell Scripting, Elm, JavaScript/TypeScript, R

Tools: AWS, GCP, Kubernetes, NumPy, SciPy, Scikit-Learn, PyTorch, Spark, Mypy, Pydantic, FastAPI, PostgreSQL, Git, LaTeX

Other: Machine Learning, Data Analysis, Linear Algebra, Bayesian/Frequentist Statistics, Simulation, CI/CD, Sensors, Analog & Digital Electronics, Scientific Computing, Neutrino & Particle Physics, Radio (Amateur Extra License), Experienced Underground Miner

Education

Insight Data Science, Mountain View, CA

- Postdoctoral Fellowship

Stanford University, Stanford, CA

- Ph.D. (Physics)

Rice University, Houston, TX

- B.S. (Physics)